Week #13: Optimization Methods

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Optimization lies at the heart of countless scientific and engineering applications, from parameter estimation and machine learning to optimal control and design problems. This week we develop the mathematical foundations of numerical optimization, exploring both unconstrained and constrained optimization methods while analyzing their convergence properties and computational characteristics.

The fundamental challenge in numerical optimization is to efficiently locate extrema of functions that may be nonlinear, non-convex, and high-dimensional. We examine how different optimization algorithms balance the competing demands of rapid convergence, global search capability, and computational efficiency, and investigate the mathematical principles that determine algorithm performance.

Our analysis emphasizes the mathematical theory underlying optimization methods, the crucial role of convexity in ensuring global convergence, and the convergence analysis that guides algorithm selection and implementation strategies.

1 Mathematical Foundations

1.1 Optimization Problem Formulation

Definition. General Optimization Problem

$$\min_{x \in \mathbb{R}^n} \quad f(x) \tag{1}$$

$$\text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \ldots, m \tag{2}$$

$$h_j(x) = 0, \quad j = 1, \ldots, p \tag{3}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, g_i are inequality constraints, and h_j are equality constraints.

Definition. Unconstrained Optimization

When $m = p = 0$, the problem reduces to:

$$\min_{x \in \mathbb{R}^n} f(x)$$

1.2 Optimality Conditions

Theorem 1.1 (First-Order Necessary Conditions)**.** *If x^* is a local minimum of f and f is differentiable at x^*, then:*

$$\nabla f(x^*) = 0$$

Theorem 1.2 (Second-Order Necessary Conditions)**.** *If x^* is a local minimum and f is twice differentiable at x^*, then:*

1. $\nabla f(x^*) = 0$
2. $\nabla^2 f(x^*)$ *is positive semidefinite*

Theorem 1.3 (Second-Order Sufficient Conditions)**.** *If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then x^* is a strict local minimum.*

1.3 Convexity

Definition. Convex Function

A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Theorem 1.4 (Global Optimality for Convex Functions)**.** *For a convex function f, any local minimum is a global minimum. If f is strictly convex, the global minimum is unique.*

Theorem 1.5 (Convexity and Second Derivatives)**.** *If f is twice differentiable, then f is convex if and only if $\nabla^2 f(x)$ is positive semidefinite for all x.*

2 Line Search Methods

Line search methods choose a direction p_k and then determine a step length α_k along that direction.

2.1 General Line Search Framework

Definition. Line Search Algorithm

1. Choose search direction p_k
2. Solve the one-dimensional problem: $\min_{\alpha>0} f(x_k + \alpha p_k)$
3. Set $x_{k+1} = x_k + \alpha_k p_k$

2.2 Wolfe Conditions

Definition. Wolfe Conditions

A step length α satisfies the Wolfe conditions if:

$$f(x_k + \alpha p_k) \leq f(x_k) + c_1 \alpha \nabla f(x_k)^T p_k \quad \text{(Armijo condition)} \tag{4}$$

$$\nabla f(x_k + \alpha p_k)^T p_k \geq c_2 \nabla f(x_k)^T p_k \quad \text{(curvature condition)} \tag{5}$$

where $0 < c_1 < c_2 < 1$.

Theorem 2.1 (Existence of Wolfe Step)**.** *For any descent direction p_k (i.e., $\nabla f(x_k)^T p_k < 0$), there exists a step length satisfying the Wolfe conditions.*

2.3 Backtracking Line Search

Definition. Backtracking Algorithm

1. Choose $\alpha_0 > 0$, $\rho \in (0, 1)$, $c \in (0, 1)$
2. Set $\alpha = \alpha_0$
3. While $f(x_k + \alpha p_k) > f(x_k) + c\alpha \nabla f(x_k)^T p_k$:
   - Set $\alpha = \rho\alpha$
4. Return $\alpha_k = \alpha$

3 Gradient-Based Methods

3.1 Steepest Descent

Definition. Steepest Descent Method

$$p_k = -\nabla f(x_k) \tag{6}$$
$$x_{k+1} = x_k + \alpha_k p_k \tag{7}$$

where α_k is chosen by line search.

Theorem 3.1 (Steepest Descent Convergence)**.** *For a convex quadratic function $f(x) = \frac{1}{2}x^T Ax - b^T x$, steepest descent with exact line search converges with:*

$$f(x_k) - f(x^*) \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^{2k} [f(x_0) - f(x^*)]$$

where κ is the condition number of A.

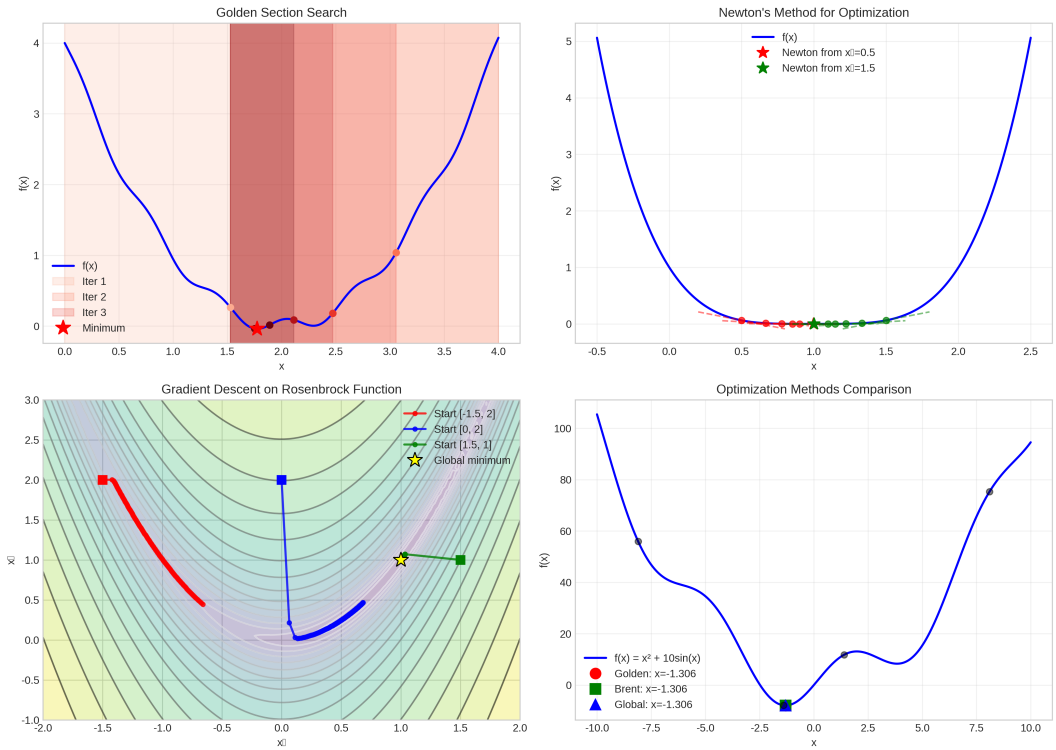

Figure 1: Optimization methods comparison. The left panel shows convergence paths for different algorithms on a quadratic function. The right panel illustrates the effect of conditioning on convergence rates.

3.2 Newton's Method

Definition. Newton's Method for Optimization

$$p_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \tag{8}$$

$$x_{k+1} = x_k + \alpha_k p_k \tag{9}$$

Theorem 3.2 (Newton's Method Convergence)**.** *If f is twice continuously differentiable, $\nabla^2 f(x^*)$ is positive definite, and x_0 is sufficiently close to x^*, then Newton's method converges quadratically.*

3.3 Quasi-Newton Methods

Quasi-Newton methods approximate the Hessian to avoid expensive second derivative computations.

Definition. Secant Equation

The quasi-Newton approximation B_{k+1} should satisfy:

$$B_{k+1} s_k = y_k$$

where $s_k = x_{k+1} - x_k$ and $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$.

Definition. BFGS Update

The Broyden-Fletcher-Goldfarb-Shanno (BFGS) update is:

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k}$$

Theorem 3.3 (BFGS Convergence)**.** *For a convex quadratic function, BFGS with exact line search converges in at most n steps, where n is the dimension.*

4 Trust Region Methods

Trust region methods choose both direction and step length by solving a subproblem within a trusted region.

4.1 Trust Region Framework

Definition. Trust Region Subproblem

At each iteration, solve:

$$\min_{p \in \mathbb{R}^n} \quad m_k(p) = f(x_k) + \nabla f(x_k)^T p + \frac{1}{2} p^T B_k p \tag{10}$$

$$\text{subject to} \quad \|p\| \leq \Delta_k \tag{11}$$

where m_k is a quadratic model and Δ_k is the trust region radius.

Definition. Trust Region Algorithm

1. Solve the trust region subproblem for p_k
2. Compute the ratio $\rho_k = \frac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$
3. If $\rho_k > \eta_1$, accept the step: $x_{k+1} = x_k + p_k$
4. Update trust region radius based on ρ_k

4.2 Cauchy Point

Definition. Cauchy Point

The Cauchy point is the minimizer of the quadratic model along the steepest descent direction within the trust region:

$$p_k^C = -\tau_k \frac{\Delta_k}{\|\nabla f(x_k)\|} \nabla f(x_k)$$

where τ_k is chosen to minimize m_k.

Theorem 4.1 (Cauchy Point Reduction)**.** *The Cauchy point achieves at least the reduction:*

$$m_k(0) - m_k(p_k^C) \geq \frac{1}{2} \|\nabla f(x_k)\| \min \left\{ \Delta_k, \frac{\|\nabla f(x_k)\|}{\|B_k\|} \right\}$$

5 Conjugate Gradient Methods

5.1 Linear Conjugate Gradient

Definition. Conjugate Directions

Directions $p_0, p_1, \ldots, p_{k-1}$ are conjugate with respect to positive definite matrix A if:

$$p_i^T A p_j = 0 \quad \text{for } i \neq j$$

Theorem 5.1 (Conjugate Direction Convergence)**.** *If we minimize a quadratic function along conjugate directions, the method converges in at most n steps.*

Definition. Conjugate Gradient Algorithm

$$r_0 = -\nabla f(x_0), \quad p_0 = r_0 \tag{12}$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k} \tag{13}$$

$$x_{k+1} = x_k + \alpha_k p_k \tag{14}$$

$$r_{k+1} = r_k - \alpha_k A p_k \tag{15}$$

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k} \tag{16}$$

$$p_{k+1} = r_{k+1} + \beta_k p_k \tag{17}$$

5.2 Nonlinear Conjugate Gradient

Definition. Fletcher-Reeves Formula

$$\beta_k^{FR} = \frac{\|\nabla f(x_{k+1})\|^2}{\|\nabla f(x_k)\|^2}$$

Definition. Polak-Ribière Formula

$$\beta_k^{PR} = \frac{\nabla f(x_{k+1})^T [\nabla f(x_{k+1}) - \nabla f(x_k)]}{\|\nabla f(x_k)\|^2}$$

Intuition: Why Conjugate Gradient Works

Conjugate gradient methods build up information about the curvature of the function through the conjugacy relationships. This allows them to achieve superlinear convergence while using only first-order information and minimal storage.

6 Constrained Optimization

6.1 Lagrange Multipliers

Theorem 6.1 (Karush-Kuhn-Tucker (KKT) Conditions)**.** *For the constrained problem with x^* a local minimum, there exist multipliers $\lambda_i \geq 0$ and μ_j such that:*

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^{p} \mu_j \nabla h_j(x^*) = 0 \tag{18}$$

$$g_i(x^*) \leq 0, \quad i = 1, \ldots, m \tag{19}$$

$$h_j(x^*) = 0, \quad j = 1, \ldots, p \tag{20}$$

$$\lambda_i \geq 0, \quad i = 1, \ldots, m \tag{21}$$

$$\lambda_i g_i(x^*) = 0, \quad i = 1, \ldots, m \tag{22}$$

6.2 Penalty Methods

Definition. Quadratic Penalty Function

$$P(x, \mu) = f(x) + \frac{\mu}{2} \sum_{j=1}^{p} [h_j(x)]^2 + \frac{\mu}{2} \sum_{i=1}^{m} [\max(0, g_i(x))]^2$$

Theorem 6.2 (Penalty Method Convergence)**.** *As $\mu \to \infty$, the minimizers of $P(x, \mu)$ converge to the solution of the constrained problem.*

6.3 Barrier Methods

Definition. Logarithmic Barrier Function

$$B(x, \mu) = f(x) - \mu \sum_{i=1}^{m} \log(-g_i(x))$$

for x in the interior of the feasible region.

7 Global Optimization

7.1 Simulated Annealing

Definition. Simulated Annealing Algorithm

1. Generate candidate x_{new} from current point $x_{current}$
2. If $f(x_{new}) < f(x_{current})$, accept x_{new}
3. Otherwise, accept x_{new} with probability $\exp(-\Delta f/T)$
4. Decrease temperature T according to cooling schedule

7.2 Genetic Algorithms

Definition. Genetic Algorithm Components

- **Population**: Set of candidate solutions
- **Selection**: Choose parents based on fitness
- **Crossover**: Combine parent solutions
- **Mutation**: Random modifications
- **Replacement**: Form new generation

8 Derivative-Free Optimization

8.1 Nelder-Mead Simplex Method

Definition. Simplex Operations
For a simplex with vertices $x_1, \ldots, x_{n+1}$:

- **Reflection**: $x_r = x_c + \alpha(x_c - x_h)$
- **Expansion**: $x_e = x_c + \gamma(x_r - x_c)$
- **Contraction**: $x_{con} = x_c + \beta(x_h - x_c)$
- **Shrinkage**: Move all vertices toward best point

where x_c is the centroid and x_h is the worst vertex.

8.2 Pattern Search Methods

Definition. Pattern Search Algorithm

1. Evaluate function at current point plus pattern vectors
2. If improvement found, move to best point and possibly expand pattern
3. Otherwise, contract pattern and stay at current point

9 Optimization in Machine Learning

9.1 Stochastic Gradient Descent

Definition. SGD Algorithm
For objective $f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$:

$$x_{k+1} = x_k - \alpha_k \nabla f_{i_k}(x_k)$$

where i_k is randomly selected.

9.2 Adam Optimizer

Definition. Adam Algorithm

$$m_k = \beta_1 m_{k-1} + (1-\beta_1)\nabla f(x_k) \tag{23}$$

$$v_k = \beta_2 v_{k-1} + (1-\beta_2)[\nabla f(x_k)]^2 \tag{24}$$

$$\hat{m}_k = \frac{m_k}{1-\beta_1^k}, \quad \hat{v}_k = \frac{v_k}{1-\beta_2^k} \tag{25}$$

$$x_{k+1} = x_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon} \tag{26}$$

Practice Problems

1. Prove that the steepest descent direction is the direction of maximum decrease for the function.
2. Implement the BFGS method and compare its performance to steepest descent on the Rosenbrock function.
3. Derive the conjugate gradient algorithm for quadratic functions from the conjugacy conditions.
4. Analyze the convergence rate of Newton's method for optimization and explain why it's quadratic.
5. Implement a trust region method and compare it to line search methods on ill-conditioned problems.

Solution:

Problem 1: The directional derivative in direction p is $\nabla f(x)^T p = \|\nabla f(x)\|\|p\| \cos\theta$. This is minimized when $\cos\theta = -1$, i.e., $p = -\nabla f(x)/\|\nabla f(x)\|$.

Problem 2: BFGS typically converges much faster than steepest descent, especially on ill-conditioned problems, because it builds up curvature information through the quasi-Newton approximation.

Problem 3: The conjugacy conditions $p_i^T A p_j = 0$ ensure that minimization along one direction doesn't spoil the minimization along previous directions, leading to finite termination.

Problem 4: Newton's method uses the exact Hessian, so the quadratic model perfectly matches the function near the optimum. This leads to quadratic convergence: $\|x_{k+1} - x^*\| \leq C\|x_k - x^*\|^2$.

Problem 5: Trust region methods are more robust than line search methods because they limit the step size when the quadratic model is unreliable, leading to better global convergence properties.